HEALTH SERVICES ADMINISTRATION PORTFOLIO · HEALTHCARE FINANCIAL ACCOUNTING

Bad Debt and Charity Care under US GAAP

A Riverside Community Hospital case study with an AI overlay

$200.0M to $61.2M
Gross charges to net patient service revenue, 30.6% of charges

$5.31M
Uncompensated care at cost, 8.6% of operating expenses

$3.1M
Operating income, a 4.8% margin

23.2% to 9.3%
Operating income at risk from a collection estimate miss, with AI segmentation (assumed)

All hospital data are simulated for academic purposes. The standards cited are real. October 2026.

COMPILED PORTFOLIO

Five deliverables, one case

This page compiles the case study, the full working analysis, the executive deck preview and the Excel model summary. The editable originals are linked below.

1	**Portfolio case study** Five-part summary written for healthcare recruiters. Sections 1 to 5 of this page.	Editable doc
2	**Working analysis** Classification, journal entries, rollforward, statements, disclosures, ratios, GAAP mapping and AI scenario. Appendix A.	Editable doc
3	**Executive deck** Twelve slides with speaker notes. Preview in the Executive deck section.	Open the deck
4	**Excel financial model** Eight sheets with live formulas and an AI on/off toggle. Summary in Appendix B.	Separate .xlsx file
5	**PDF copy of this page** Print layout of everything on this page, including the deck preview.	Separate PDF file

1 EXECUTIVE SUMMARY AND PROBLEM STATEMENT

AI sharpens how a hospital reports uncompensated care

> In this simulation, a 250-bed nonprofit hospital reports $5.31M of uncompensated care correctly under US GAAP, and AI screening and segmented estimates cut its collection-estimate risk from 23.2% to 9.3% of operating income.

Problem statement. Hospitals often blur three different things: charity care (care never expected to be paid), implicit price concessions (an expected collection shortfall) and credit losses (a payer or patient defaulting after the price was set). Each has a different GAAP treatment. When they are mixed, revenue is overstated and later reversed, charity care is understated in the notes and community benefit reporting, and auditors find classification errors.

Approach. The project builds a fictional hospital, Riverside Community Hospital, with $200M of annual gross charges, then:

- classifies every dollar of non-payment under ASC 606, ASC 954-605 and ASC 326;
- records the full year in journal entries, an allowance rollforward and a statement of operations;
- adds an AI layer for charity screening and collection estimates, with the governance an auditor would expect.

Headline results (simulated):

- $200.0M of gross charges becomes $61.2M of net patient service revenue (30.6%).
- Uncompensated care at cost is $5.31M, or 8.6% of operating expenses.
- AI screening at registration moves $3M of charges from bad debt to charity care, lifting reported charity care at cost from $1.8M to $2.7M.
- Segmenting accounts by propensity to pay cuts the revenue true-up risk from $720K to $288K, on the stated assumptions.

Skills shown: healthcare revenue cycle accounting, regulatory compliance, financial modeling, estimate governance and responsible AI.

All hospital data are simulated for academic purposes. The standards cited are real, and the case was prepared as a Health Services Administration coursework project in Healthcare Financial Accounting.

2 METHODOLOGY AND REGULATORY FRAMEWORKS APPLIED

Six frameworks, applied to one simulated year

The analysis applies six accounting, regulatory and costing frameworks to one

simulated year, then layers AI governance on top.

FRAMEWORK	ROLE IN THE PROJECT	WHERE IT SHOWS UP
ASC 606, Revenue from Contracts with Customers	Expected collection shortfalls are variable consideration; implicit price concessions reduce the transaction price	Gross-to-net schedule; journal entries 3 to 5
ASC 954-605, Health Care Entities, Revenue	Charity care is not revenue or a receivable; policy and cost are disclosed (954-605-25-10, 954-605-50-3)	Charity care entry; footnote
ASC 326, Credit Losses	A payer or patient default from credit deterioration is an operating expense	$400K insurer credit loss
HFMA Statement 15 (June 2019)	Separates charity care, implicit price concessions and bad debt; sets timing, reclassification and documentation practice	Retroactive charity reclassification; classification table
IRS Section 501(r) and Form 990 Schedule H	Financial assistance policy and community benefit reporting for tax-exempt hospitals	Policy assumptions; referenced, not modeled in detail
Medicare cost report cost-to-charge ratio	Basis for costing charity care at 30% of charges (assumed)	$1.8M charity care at cost
Model risk and AI governance practice	Validation, back-testing, human review, privacy	Intervention 5

Method

1. Define the categories and the GAAP basis for each.
2. Build the case: payer mix, contractual rates, policy discounts, collection history and cost-to-charge ratio.
3. Record the year in journal entries and an allowance rollforward.
4. Present the statement of operations and the charity care disclosure.
5. Overlay AI screening and propensity-to-pay segmentation, and measure the change in classification and estimate risk.
6. Stress test the collection assumption and document controls.

Limits. The data are simulated for one hospital with zero opening balances. Medicare bad debt reimbursement and sliding-scale discounts are excluded, and the AI effects rest on stated assumptions, not observed results.

3 MOCK DATA AND STRATEGIC INTERVENTIONS

A 4.8% margin, with 8.6% of expense going to uncompensated care

Riverside starts the year at a 4.8% operating margin, with 8.6% of its operating expense going to uncompensated care, and six interventions target how that care

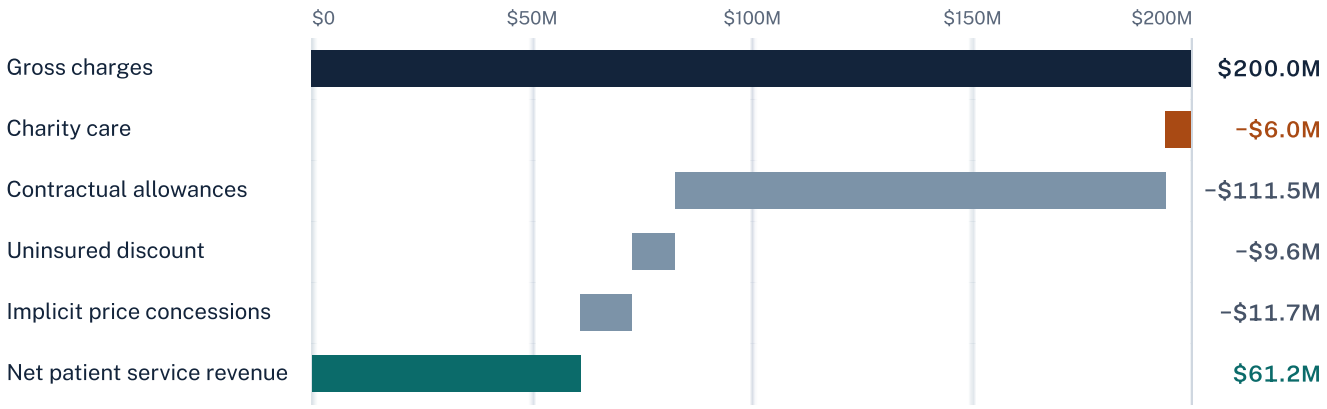
is classified and estimated.

Baseline, Riverside Community Hospital, FY2025 (simulated)

METRIC	VALUE
Gross patient service charges	$200.0M
Net patient service revenue	$61.2M (30.6% of charges)
Implicit price concessions	$11.7M (5.85% of charges)
Charity care	$6.0M of charges; $1.8M at cost (2.9% of operating expenses)
Uncompensated care at cost	$5.31M (8.6% of operating expenses)
Provision for credit losses	$0.4M (insurer in receivership)
Net patient accounts receivable	$3.0M (17.9 days)
Operating income	**$3.1M (4.8% margin)**

Key assumptions: self-pay patients collect 25% of billed amounts after a 40% uninsured discount, insured patient balances collect 80%, and cost is 30% of charges.

Where $200M of charges goes



Each bar sits where its step lands on one $0 to $200M scale. Charity care leaves revenue entirely and appears here only to bridge from the chargemaster. Simulated FY2025 data.

Strategic interventions

#	INTERVENTION	WHAT CHANGES	OWNER	ANCHOR
1	**Written classification policy**	A decision tree separates charity care, implicit price concession and credit loss, approved by the governing body	CFO and revenue cycle VP	ASC 954-605; HFMA Statement 15, section 3.2
2	**GAAP-compliant recording model**	Separate contra revenue accounts for charity, contractual, uninsured discount and implicit price concessions; separate allowances for concessions and credit losses	Controller	ASC 606; ASC 326
3	**AI presumptive charity screening**	Scores patients at registration, flags likely eligible accounts before the first bill, and routes them to human review	Patient access director	Statement 15, section 4.1
4	**Segmented concession estimate with back-testing**	Estimates the concession by high, medium and low propensity segments and back-tests it against cash every quarter	Revenue cycle analytics and Controller	ASC 606; Statement 15, sections 8.1(d) and 8.3
5	**AI governance and controls**	Model validation, fairness testing, audit trail, and exception reports on write-offs and reclassifications	Compliance and internal audit	Statement 15, section 5 recordkeeping; HIPAA
6	**Disclosure package**	Charity care note at cost plus a supplemental uncompensated care table	Financial reporting	ASC 954-605-50-3

The AI assumptions are that screening flags $3M of self-pay charges as charity-eligible at registration, and that segmentation reduces the typical collection-rate miss from 5 points to 2.

4 KEY DELIVERABLES AND OUTCOMES

Reported charity care rises 50%, true-up risk falls by more than half

The interventions make reported charity care 50% higher and cut the revenue true-up risk by more than half, at the cost of $450K of revenue that was never collectible.

OUTCOME MEASURE	BASELINE	WITH INTERVENTIONS	BASIS
Charity care at cost	$1.80M	**$2.70M**	Modeled
Implicit price concessions at cost	$3.51M	$3.105M	Modeled
Uncompensated care at cost	$5.31M	$5.805M	Modeled
Net patient service revenue	$61.20M	$60.75M	Modeled
Typical collection-rate miss	5 points	2 points	Assumed
Revenue true-up risk (share of operating income)	$720K (23.2%)	**$288K (9.3%)**	Derived from the assumed miss
Charity determinations made before the first statement	35%	80%	Mock target, not modeled

Two readings matter to an executive audience. First, reported operating income falls from $3.1M to $2.65M in the screening scenario, because the $450K of expected collections on those accounts was never realistic. The gain is accuracy and earlier recognition, not extra profit. Second, the mix shifts from bad debt toward charity care, which strengthens community benefit and tax-exemption reporting.

Deliverables produced

Full workings are in Appendix A.

- GAAP classification table for charity care, implicit price concessions and credit losses
- Eight journal entries plus a retroactive charity care reclassification
- Allowance rollforward and year-end receivables schedule
- Statement of operations and the charity care note at cost
- Ratio analysis and a collection-rate sensitivity
- AI screening and segmentation scenario with governance guardrails

Recommendation to leadership. Adopt the classification policy and recording model first, because they change no cash and fix the reporting. Pilot AI screening on one entry point in stages, back-test it for two quarters, and extend only once the variance stays inside the agreed threshold.

5 VISUAL ASSETS AND ARTIFACTS

Nine assets turn the analysis into portfolio evidence

The first four, an Excel model, a waterfall chart, a decision tree and an executive deck, carry most of the weight with recruiters. Three are built and the rest are listed in build order.

ORDER	ASSET	WHAT IT SHOWS	FORMAT	STATUS
1	**Financial model workbook**	Assumptions, journal entries, allowance rollforward, statement of operations and a sensitivity, with live formulas and an AI on/off scenario toggle	Excel	BUILT
2	**Gross-to-net waterfall chart**	$200M of charges falling to $61.2M of net revenue, step by step	Chart on this page and deck slide 6	BUILT
3	**Classification decision tree**	One page that sorts a balance into charity care, implicit price concession or credit loss	PDF or diagram	TO BUILD
4	**Executive slide deck** (12 slides)	Problem, three categories, frameworks, baseline, waterfall, AI overlay, outcomes, governance, recommendation	Slides, exported to PDF	BUILT
5	**Uncompensated care dashboard mockup**	Tiles for net revenue, charity care at cost, concessions as % of charges, uncompensated care as % of expense and days in AR; a quarterly back-test of forecast against cash by segment	Design mockup	TO BUILD
6	**Patient financial clearance flow**	Registration, AI screening, human review, then charity care or billing	Diagram	TO BUILD
7	**Charity care note and policy excerpt**	The sample footnote plus a one-page financial assistance policy summary	Document	TO BUILD
8	**AI governance one-pager**	Model validation, fairness testing, approval gates and audit trail checklist	Document	TO BUILD
9	**Portfolio abstract**	A 150-word summary and a short LinkedIn post linking the deck and workbook	Text	TO BUILD

Presentation tips for recruiters

- State on every asset that the data are simulated.
- Lead each asset with its one-sentence finding, such as “AI screening moves $3M from bad debt to charity care”.
- Keep each document to one page where possible and export it as PDF.
- Add a short “what I would do differently” note on limits, which signals judgment.

EXECUTIVE DECK

Twelve slides, from problem to recommendation

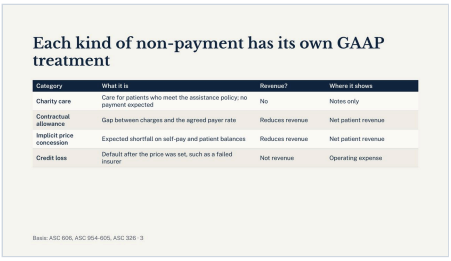
The deck walks a hospital leadership team from the three categories of non-payment to the AI pilot recommendation. Each slide has speaker notes in the live version.



1 Title



2 The problem



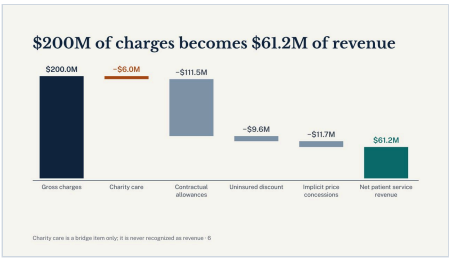
3 Three categories, three treatments



4 Four sources of guidance



5 Riverside baseline



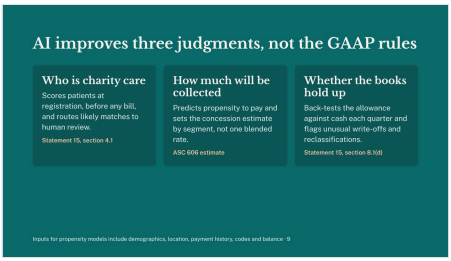
6 $200M of charges becomes $61.2M



7 Eight journal entries



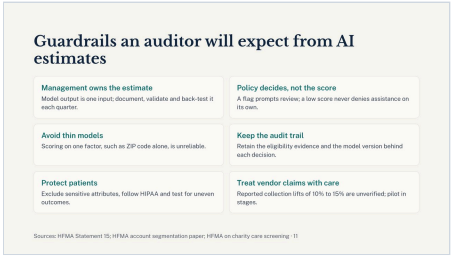
8 Statement of operations



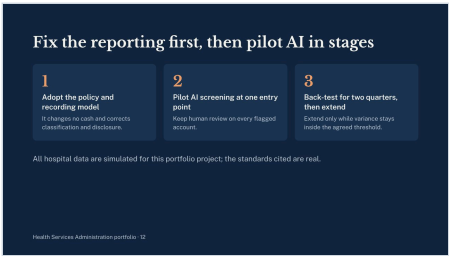
9 AI improves three judgments



10 AI scenario results



11 Guardrails



12 Recommendation

[Open the live deck]

APPENDIX A

Working analysis

The full accounting workings behind the case study, in the order a reviewer would check them.

A1. Four ways a patient balance leaves revenue

The label chosen for an uncollected balance decides which financial statement line it hits and whether it appears in the charity care disclosure.

ITEM	WHAT IT IS	REVENUE?	RECEIVABLE?	WHERE IT SHOWS
Charity care	Care given free or at a reduced charge to patients who meet the hospital's financial assistance policy; no payment expected	No	No	Excluded from revenue and AR; disclosed in the notes at cost
Contractual allowance (explicit price concession)	Gap between gross charges and the rate agreed with Medicare, Medicaid or a commercial payer	Reduces gross charges	Reduces gross AR	Gross-to-net deduction; net patient service revenue
Implicit price concession	Amount the hospital expects not to collect from self-pay or patient-responsibility balances, based on history, even though it never formally agreed to a discount	Reduces transaction price	Reduces AR through an allowance	Deduction from net patient service revenue
Credit loss (bad debt expense)	Loss from a payer or patient credit deterioration after the price was set	Not a revenue item	Reduces AR through a credit loss allowance	Operating expense (provision for credit losses)

Three questions separate them. Did the patient qualify under the written assistance policy (charity care)? Did the hospital expect, at billing, to collect less than the price (implicit price concession)? Did the expected collection fall because the payer or patient's credit worsened later (credit loss)?

A2. Accounting rules that drive the entries

Since the adoption of ASC 606, most patient non-payment lowers revenue directly, and only true credit deterioration is a separate expense.

- **Charity care.** ASC 954-605-25-10 says charity care does not qualify for recognition as revenue or as a receivable, and only the part of an account meeting the hospital's own criteria counts as charity (PwC Health Care guide).
- **Disclosure.** ASC 954-605-50-3 requires the policy and the level of charity care, measured at direct and indirect cost, with the estimation method disclosed. A cost-to-gross-charges ratio applied to charity charges is one accepted technique.
- **Implicit price concessions.** Under ASC 606, an expected collection shortfall is variable consideration that lowers the transaction price, so most amounts once shown as bad debt expense are netted inside patient service revenue (HFMA Statement 15, section 8.1).
- **Credit losses.** When a later default comes from credit deterioration rather than an expected price concession, the loss is an impairment (credit loss, ASC 326) reported as bad debt within expenses (Statement 15, section 8.1(a)).
- **Classification guidance.** HFMA Statement 15 (June 2019) adds implicit price concessions as a third category beside charity care and bad debt within uncompensated care, and stresses keeping the three separate.

A tax-exempt hospital also follows its Section 501(r) financial assistance policy, which feeds Form 990 Schedule H. The case below excludes Medicare bad debt reimbursement for deductibles and coinsurance, which flows through the cost report.

A3. Case facts: Riverside Community Hospital, FY2025

Riverside Community Hospital is a fictional 250-bed nonprofit acute care hospital with a December 31 year-end. In FY2025 it billed $200,000,000 of gross charges from its chargemaster.

PAYER	GROSS CHARGES ($)	CONTRACTUAL ADJUSTMENT ($)	EXPECTED PAYER AND PATIENT AMOUNT ($)
Medicare	70,000,000	49,000,000 (70%)	21,000,000
Medicaid	30,000,000	24,000,000 (80%)	6,000,000
Commercial	70,000,000	38,500,000 (55%)	31,500,000, of which patient copays and deductibles are 4,500,000
Self-pay (uninsured)	30,000,000	none	see policy assumptions
Total	**200,000,000**	**111,500,000**	

The policy and estimation assumptions used throughout:

1. **Charity care.** Self-pay patients at or below 200% of the federal poverty level receive free care under the written financial assistance policy. Eligibility was determined at or before billing for $6,000,000 of gross charges.
2. **Uninsured discount.** The remaining $24,000,000 of self-pay charges receives a 40% policy discount ($9,600,000), leaving $14,400,000 billed to patients.
3. **Self-pay collection history.** Historical experience is that 25% of billed self-pay amounts are collected, so the implicit price concession is 75%, or $10,800,000.
4. **Insured patient responsibility.** Commercial copays and deductibles of $4,500,000 collect at 80%, so the implicit price concession is $900,000.
5. **Cost-to-charge ratio.** The Medicare cost report supports a ratio of 30% of operating cost to gross charges, used to cost charity care.
6. **Credit event.** In the fourth quarter a commercial insurer enters receivership while owing $1,000,000; management reserves 40%, or $400,000, as a credit loss.
7. **Cash and write-offs.** Cash collected is $57,800,000 and accounts written off to collections total $9,700,000, detailed in the journal entries.

A4. From gross charges to net patient service revenue

$200,000,000 of gross charges becomes $61,200,000 of net patient service revenue, or 30.6% of charges, once charity care, contractual allowances, discounts and implicit price concessions come out.

LINE	AMOUNT ($)
Gross patient service charges	200,000,000
Less: charity care (charges foregone)	(6,000,000)
Less: contractual allowances (Medicare, Medicaid, commercial)	(111,500,000)
Less: uninsured self-pay discount	(9,600,000)
Amounts expected from payers and patients before collection shortfall	**72,900,000**
Less: implicit price concession, self-pay	(10,800,000)
Less: implicit price concession, insured patient responsibility	(900,000)
Net patient service revenue	**61,200,000**

Charity care is shown here only to bridge from the chargemaster. The published statements start at net patient service revenue and never report charity care as revenue; the cost of it goes in the notes (A8). The credit loss on the failed insurer is not in this schedule because it is an expense, not a price adjustment.

A5. Journal entries

Eight summary entries take the year from billing to write-off, and entries 1 through 5 and 8 touch the operating statement while entries 6 and 7 do not.

JE	EVENT	ACCOUNT	DEBIT ($)	CREDIT ($)
1	Record gross charges	`Patient accounts receivable (gross)`	200,000,000	
		`  Gross patient service revenue`		200,000,000
2	Charity eligibility determined	`Charity care (contra revenue)`	6,000,000	
		`  Patient accounts receivable`		6,000,000
3	Payer contracts	`Contractual allowances (contra revenue)`	111,500,000	
		`  Patient accounts receivable`		111,500,000
4	Uninsured policy discount	`Uninsured discounts (contra revenue)`	9,600,000	
		`  Patient accounts receivable`		9,600,000
5	Estimate implicit price concessions	`Implicit price concessions (contra revenue)`	11,700,000	
		`  Allowance for implicit price concessions`		11,700,000
6	Cash collected	`Cash`	57,800,000	
		`  Patient accounts receivable`		57,800,000
7	Write off accounts sent to collections	`Allowance for implicit price concessions`	9,700,000	
		`  Patient accounts receivable`		9,700,000
8	Insurer receivership, 40% of $1,000,000	`Provision for credit losses (operating expense)`	400,000	
		`  Allowance for credit losses`		400,000

Support for entries 6 and 7

SOURCE	CASH COLLECTED ($)	WRITTEN OFF ($)
Medicare	20,000,000	
Medicaid	5,500,000	
Commercial payer	26,000,000	
Commercial patient responsibility	3,300,000	700,000
Self-pay	3,000,000	9,000,000
Total	**57,800,000**	**9,700,000**

Entry 7 has no income statement effect. The reduction in revenue was booked in entry 5, so a write-off only moves a balance out of AR and its allowance.

Retroactive charity care (illustration, outside the totals above). A patient with $50,000 of charges was billed as self-pay: a 40% discount ($20,000) left $30,000 in AR, and $22,500 of it sat in the implicit price concession allowance. The patient then applies for assistance inside the policy's application window and qualifies for free care. The hospital reclassifies the account instead of writing off bad debt:

ACCOUNT	DEBIT ($)	CREDIT ($)
Allowance for implicit price concessions	22,500	
Charity care (contra revenue)	27,500	
Uninsured discounts (reverse)		20,000
Patient accounts receivable		30,000

The full $50,000 of charges now sits in charity care, so net revenue is unchanged, and the account leaves AR and the uncompensated care figure moves from bad debt to charity care. HFMA Statement 15 supports this: section 8.5 calls for reclassifying a bad debt to charity care when collection work shows the patient was eligible, and section 4.2 says a policy should define how long after service a patient can qualify. The Section 501(r) final regulations also set a minimum application period (240 days after the first post-discharge bill, from memory and not reviewed for this project).

A6. Allowance rollforward and year-end receivables

Net patient accounts receivable ends FY2025 at $3,000,000 after $2,400,000 of allowances, and the patient balances still on the books are reserved at 69%. Opening balances are assumed to be zero to isolate the year's activity.

ALLOWANCE ROLLFORWARD ($)	IMPLICIT PRICE CONCESSIONS	CREDIT LOSSES
Beginning balance	0	0
Provision (JE 5, JE 8)	11,700,000	400,000
Write-offs (JE 7)	(9,700,000)	0
Ending balance	**2,000,000**	**400,000**

YEAR-END RECEIVABLES ($)	GROSS AR	EXPECTED TO COLLECT
Medicare	1,000,000	1,000,000
Medicaid	500,000	500,000
Commercial payer (insurer in receivership)	1,000,000	600,000
Commercial patient responsibility	500,000	300,000
Self-pay	2,400,000	600,000
Total	**5,400,000**	**3,000,000**
Less: allowance for implicit price concessions	(2,000,000)	
Less: allowance for credit losses	(400,000)	
Net patient accounts receivable	**3,000,000**	

The allowance for implicit price concessions equals the gap on patient balances: self-pay $1,800,000 plus insured patient responsibility $200,000. The credit loss allowance is held

separately because it answers a different question: whether the insurer will pay, not what price the hospital expects.

A7. Statement of operations presentation

Riverside reports operating income of $3,100,000, a 4.8% margin, with charity care absent from revenue and the only bad debt-type expense being the $400,000 credit loss.

STATEMENT OF OPERATIONS, YEAR ENDED DECEMBER 31, 2025	$
Net patient service revenue (net of charity care, contractual allowances, discounts and implicit price concessions)	61,200,000
Other operating revenue	3,800,000
Total operating revenue	**65,000,000**
Salaries and benefits	31,000,000
Supplies and drugs	11,500,000
Purchased services	6,200,000
Depreciation	4,100,000
Interest	1,400,000
Other operating expenses	7,300,000
Provision for credit losses	400,000
Total operating expenses	**61,900,000**
Operating income	**3,100,000**

Before ASC 606, a hospital showed its whole provision for bad debts, here $12,100,000 ($11,700,000 of patient non-payment plus the $400,000 credit loss), as a deduction in the revenue section. Under ASC 606 the $11,700,000 is inside net patient service revenue and only the $400,000 is a separate line, so operating income is the same but the reported revenue line and the bad debt line are not comparable across the adoption date.

A8. Disclosures

Riverside discloses $1,800,000 of charity care at cost, and shows the wider uncompensated care picture of $5,310,000 as supplemental information.

The cost of charity care is the gross charges foregone times the cost-to-charge ratio: $6,000,000 × 30% = $1,800,000. This follows the estimation method ASC 954-605-50-3 describes when costs cannot be traced to charity patients directly. Sample note wording:

> **Charity care.** Riverside provides care without charge to patients whose household income is at or below 200% of the federal poverty level and who meet the requirements of its financial assistance policy. Because Riverside does not expect payment, charity care is not reported as revenue or receivables. The estimated cost of charity care was $1,800,000 for 2025, determined by applying a ratio of cost to gross charges to the gross charges for charity patients.

The note covers the required items: the policy, the level of charity care, and the cost method. The table below is supplemental (for the MD&A, a board report or Form 990 Schedule H support), not a GAAP-required note, and it follows the three HFMA categories.

UNCOMPENSATED CARE AT COST, FY2025	GROSS AMOUNT ($)	COST AT 30% ($)
Charity care	6,000,000	1,800,000
Implicit price concessions (self-pay and insured patient balances)	11,700,000	3,510,000
Total uncompensated care at cost		**5,310,000**

The $400,000 insurer credit loss is left out because it is a payer failure, not care to a patient who could not pay. The 40% uninsured discount is also not charity care, since Statement 15 (section 6.3(b)) keeps discounts outside the charity policy out of the charity care note. Charity care and bad debt are reported separately because a charity patient was never expected to pay, while a bad debt patient was billed and failed to pay; blending them would overstate collection effort and understate the community benefit.

A9. Ratios, sensitivity and controls

Uncompensated care costs Riverside 8.6% of operating expenses, and the self-pay collection estimate is the single largest swing factor in reported income.

MEASURE	CALCULATION	RESULT
Charity care at cost, % of operating expenses	1,800,000 / 61,900,000	2.9%
Uncompensated care at cost, % of operating expenses	5,310,000 / 61,900,000	8.6%
Implicit price concessions, % of gross charges	11,700,000 / 200,000,000	5.85%
Net patient service revenue, % of gross charges	61,200,000 / 200,000,000	30.6%
Allowance coverage of patient balances	2,000,000 / 2,900,000	69%
Days in net patient AR	3,000,000 / (61,200,000 / 365)	17.9 days
Operating margin	3,100,000 / 65,000,000	4.8%

Sensitivity. Each percentage point of self-pay collection rate moves revenue by $144,000, which is 4.6% of operating income. If collections run 20% instead of 25%, the implicit price concession rises by $720,000, operating income falls to $2,380,000 (down 23.2%) and the margin to 3.7%. A concession estimate is a revenue estimate, so it belongs in critical accounting estimates.

Controls and audit risks:

- **Classification date.** Document when charity eligibility was decided, because moving an account between charity care and bad debt changes both revenue disclosures and the uncompensated care report.
- **Estimate support.** Build the concession rate from historical collections by payer class and aging, and back-test it against actual cash each quarter.
- **Portfolio approach.** Apply the estimate to groups of similar accounts, and document why the portfolio result would not differ materially from account-by-account review.
- **Write-off discipline.** Write off only against the allowance and monitor allowance balances for aged accounts that were never written off.
- **Policy consistency.** Apply the financial assistance policy as written, and disclose any policy change that moves the charity care amount.
- **Presentation.** Keep credit losses separate from price concessions, and keep charity care out of revenue and receivables.

A10. US GAAP mapping of the case

Every line of the Riverside case has a specific US GAAP basis, and the basis decides whether the amount is revenue, expense or a note disclosure only.

CASE ITEM	GAAP BASIS	TREATMENT	REPORTED IN
Charity care ($6,000,000 of charges)	ASC 954-605-25-10 and 954-605-50-3	Not revenue and not a receivable; policy, level and cost method disclosed	Notes only
Contractual allowances ($111,500,000)	ASC 606, explicit price concessions in the transaction price	Reduce gross charges to the payer rate	Net patient service revenue
Uninsured discount ($9,600,000)	ASC 606 price concession; not charity care under HFMA Statement 15, section 6.3(b)	Reduces revenue	Net patient service revenue
Implicit price concessions ($11,700,000)	ASC 606 variable consideration	Reduces revenue through an allowance, evaluated regularly	Net patient service revenue; optional note
Credit loss on the insurer ($400,000)	ASC 326 and Statement 15, section 8.1(a)	Operating expense through a credit loss allowance	Operating expenses
Changes in the collection estimate	Statement 15, section 8.3	Accounted for in the period the new information arrives	Current-period revenue
Payer mix and judgments	ASC 606 disclosure objectives (Statement 15, section 9.2)	Disaggregate revenue and describe significant judgments	Notes

A11. How AI solves the problem

AI does not change the GAAP classification rules; it improves the three judgments GAAP leaves to management: who is charity care, how much of the rest will be collected, and whether the records prove it.

JUDGMENT	WHAT AI DOES	GAAP LINK
Who qualifies for charity care	Scores patients at registration for presumptive eligibility using financial and demographic data, before any bill goes out (HFMA)	Statement 15, section 4.1, asks for determinations before or at service; ASC 954-605-25-10 keeps charity out of revenue and receivables
How much self-pay and patient balances will be collected	Predicts each account's propensity to pay from demographics, location, payment history, codes and balance, and ranks accounts high, medium or low (HFMA account segmentation)	Gives a portfolio-level implicit price concession estimate under ASC 606, updated as information arrives (Statement 15, section 8.3)
Whether the allowance and classification hold up	Back-tests the estimate against actual cash each quarter and flags unusual write-offs and charity reclassifications	Statement 15, section 8.1(d), requires regular evaluation of the allowance; section 8.5 covers reclassification to charity care
Whether the disclosures are supported	Drafts the charity care note and ties the rollforward to the ledger for human review	ASC 954-605-50-3 policy, level and cost method

RIVERSIDE WITH AI SCREENING AT REGISTRATION

Assume, for illustration, that screening flags $3,000,000 of self-pay charges as charity-eligible at registration. Those accounts would otherwise have been discounted 40% ($1,200,000), billed ($1,800,000) and reserved at 75% ($1,350,000). The entry becomes a single debit to charity care and credit to AR of $3,000,000.

MEASURE	WITHOUT SCREENING ($)	WITH AI SCREENING ($)
Charity care (gross charges)	6,000,000	9,000,000
Uninsured discount	9,600,000	8,400,000
Implicit price concessions	11,700,000	10,350,000
Net patient service revenue	61,200,000	60,750,000
Charity care at cost (30%)	1,800,000	2,700,000
Implicit price concessions at cost (30%)	3,510,000	3,105,000
Uncompensated care at cost	**5,310,000**	**5,805,000**

Revenue falls $450,000 because the 25% collection ($1,800,000 × 25%) that the average rate assumed for these patients was never realistic. Without screening, that amount is booked, the accounts age to write-off, and the allowance proves too small, so the true-up hits a later period. With screening, the statements are right the first time and the charity care note reports $2,700,000 at cost instead of $1,800,000. Uncompensated care at cost rises from $5,310,000 to $5,805,000 because the screened accounts now carry their full cost as charity care, and the mix moves from bad debt toward charity care.

BETTER ESTIMATES CUT TRUE-UP RISK

From A9, each point of self-pay collection rate is worth $144,000. Segmenting accounts by propensity to pay lets management estimate the concession by segment instead of one blended 75% rate. Assuming the legacy estimate misses by 5 points and the AI-assisted one by 2 (both assumptions, to be replaced by Riverside's own back-test results):

ESTIMATE	TYPICAL MISS	REVENUE TRUE-UP ($)	SHARE OF OPERATING INCOME
Blended historical rate	5 points	720,000	23.2%
AI-segmented rate	2 points	288,000	9.3%

GUARDRAILS GAAP AND AUDITORS WILL EXPECT

- **Management owns the estimate.** The model output is one input to the ASC 606 estimate. Document its inputs, validate it, back-test it each quarter and refresh it, since models must be built, tested and refreshed over time.
- **Policy decides charity, not the score.** A flag only prompts a review. Eligibility follows the written policy approved by the governing body (Statement 15, section 3.2), and a low score should never by itself deny assistance without human review.
- **Avoid thin models.** Scoring on one factor such as credit score or ZIP code is considered unreliable, and one model may miss local differences such as urban and rural patients.
- **Keep the audit trail.** Retain the eligibility evidence and the model version behind each determination, as Statement 15 expects documentation for charity care.
- **Protect patients.** Keep sensitive attributes out of the features, follow HIPAA, and check the results for uneven outcomes across patient groups.
- **Treat vendor claims with care.** Reported collection gains of 10% to 15% come from a vendor document and are not independently verified, and some executives doubt the accuracy of these tools, so hospitals pilot them in stages and weigh cost.

APPENDIX B

Excel financial model

One workbook records the year and runs both scenarios. Change the blue inputs on the Assumptions sheet, and set cell B5 to 1 to switch on the AI screening scenario.

SHEET	WHAT IT HOLDS
README	Purpose, colour legend, sheet guide, GAAP basis and limits
Assumptions	Every input with its basis; the AI scenario switch (B5) and the AI inputs
Gross_to_Net	Payer table, self-pay build and the gross-to-net revenue schedule
Journal_Entries	Eight summary entries with a debit and credit balance check, plus cash and write-off support
Rollforward	Allowance rollforward and year-end receivables, with a tie-out check
Statement	Statement of operations, charity care disclosure at cost and ratios
Scenario_Compare	Baseline versus AI screening side by side, independent of the switch
Sensitivity	Self-pay collection rate sensitivity and the collection estimate miss comparison

Controls built into the workbook

- 206 live formulas; the file was recalculated with zero formula errors.
- Four tie-out checks (gross-to-net, debits equal credits, net AR equals expected collections, scenario revenue) all equal zero with the switch on and off.
- Baseline results match this page: $61.2M net revenue, $3.1M operating income, $5.31M uncompensated care at cost. With the switch on: $60.75M, $2.65M and $5.805M.
- Inputs are blue, formulas black, cross-sheet links green; the AI switch is yellow.

SOURCES

Standards and guidance opened for this project

The case, its figures and Riverside Community Hospital are fictional. The pages below were opened for this project.

- HFMA Principles and Practices Board Statement 15 (June 2019): categories, recordkeeping, timing, valuation and disclosure of charity care, implicit price concessions and bad debts.
- PwC Viewpoint, Health Care Guide, uncompensated services: ASC 954-605-25-10 and ASC 954-605-50-3 on charity care recognition and cost-based disclosure.
- HFMA, hospitals implement charity care screenings: automated presumptive screening at the front end, and its caveats.
- HFMA, account segmentation (September 2023): propensity-to-pay models, inputs, tiers and limits.

Check the current Codification text and your auditor's guidance before using this in a filing. The 240-day Section 501(r) application window was not reviewed for this project.

Simulated data for an academic portfolio project in Health Services Administration.
Riverside Community Hospital is fictional.